involving larger and denser clumps.

This is an important point that is sometimes overlooked. The overwhelming drive toward disorder does not mean that orderly structures like stars and planets, or orderly life forms like plants and animals, can't form. They can. And they obviously do. What the second law of thermodynamics entails is that in the formation of order there is generally a more-than-compensating generation of disorder. The entropy balance sheet is still in the black even though certain constituents have become more ordered. And of the fundamental forces of nature, gravity is the one that exploits this feature of the entropy tally to the hilt. Because gravity operates across vast distances and is universally attractive, it instigates the formation of the ordered clumps—stars—that give off the light we see in a clear night sky, all in keeping with the net balance of entropy increase.

The more squeezed, dense, and massive the clumps of gas are, the larger the overall entropy. Black holes, the most extreme form of gravitational clumping and squeezing in the universe, take this to the limit. The gravitational pull of a black hole is so strong that nothing, not even light, is able to escape, which explains why black holes are black. Thus, unlike ordinary stars, black holes stubbornly hold on to all the entropy they produce: none of it can escape the black hole's powerful gravitational grip.[16] In fact, as we will discuss in Chapter 16, nothing in the universe contains more disorder—more entropy—than a black hole.[15] This makes good intuitive sense: high entropy means that many rearrangements of the constituents of an object go unnoticed. Since we can't see inside a black hole, it is impossible for us to detect *any* rearrangement of its constituents— whatever those constituents may be —and hence black holes have maximum entropy. When gravity flexes its muscles to the limit, it becomes the most efficient generator of entropy in the known universe.

We have now come to the place where the buck finally stops. *The ultimate source of order, of low entropy, must be the big bang itself.* In its earliest moments, rather than being filled with gargantuan containers of entropy such as black holes, as we would expect from probabilistic considerations, for some reason the nascent universe was filled with a hot, uniform, gaseous mixture of hydrogen and helium. Although this